

Partial Differential Equations for Finance

1 Stochastic Calculus Foundations

1.1 The Stochastic Differential Equation (SDE) Model

The state (e.g. price) y_t of an asset is governed by

$$dy_t = F(y_t, t) dt + G(y_t, t) dW_t,$$

where W_t is one-dimensional standard Brownian motion. The canonical example is **geometric Brownian motion** (GBM) :

$$F(x, t) = \mu x, \quad G(x, t) = \sigma x \quad \implies \quad dy_t = \mu y_t dt + \sigma y_t dW_t.$$

1.2 Itô's Formula

For a smooth function $u(x, t)$ and the SDE above,

$$du(y_s, s) = \left(F(y_s, s) \frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} + \frac{G(y_s, s)^2}{2} \frac{\partial^2 u}{\partial x^2} \right) ds + G(y_s, s) \frac{\partial u}{\partial x} dW_s.$$

Key property used throughout : The expectation of a stochastic integral vanishes, $\mathbb{E} \left[\int_t^T f(y_s, s) dW_s \right] = 0$ for any adapted process.

1.3 From SDE to PDE : The Kolmogorov Backward Equation

Basic expected payoff. Define $u(x, t) = \mathbb{E}_{y_t=x}[\Phi(y_T)]$. Applying Itô's formula and taking expectations yields the PDE

$$\frac{\partial u}{\partial t} + F \frac{\partial u}{\partial x} + \frac{G^2}{2} \frac{\partial^2 u}{\partial x^2} = 0, \quad u(x, T) = \Phi(x).$$

Technique (Sheet 1, Q1) : Write

$$0 = \mathbb{E}_{y_t=x}[u(y_T, T) - u(y_t, t)] = \mathbb{E} \left[\int_t^T du(y_r, r) \right],$$

expand via Itô and set the dr-coefficient to zero.

With discounting (Sheet 1, Q1). $u(x, t) = \mathbb{E}_{y_t=x} \left[e^{-\int_t^T b(y_s, s) ds} \Phi(y_T) \right]$ satisfies

$$\frac{\partial u}{\partial t} + F \frac{\partial u}{\partial x} + \frac{G^2}{2} \frac{\partial^2 u}{\partial x^2} - b u = 0, \quad u(x, T) = \Phi(x).$$

With running payoff (Sheet 1, Q2).

$$u(x, t) = \mathbb{E}_{y_t=x} \left[\int_t^T \Psi(y_s, s) ds + \Phi(y_T) \right] \text{ satisfies}$$

$$\frac{\partial u}{\partial t} + F \frac{\partial u}{\partial x} + \frac{G^2}{2} \frac{\partial^2 u}{\partial x^2} + \Psi = 0, \quad u(x, T) = \Phi(x).$$

Knockout option (Sheet 1, Q2). For $\tau(x)$ = first exit time from (a, b) , $u(x, t) = \mathbb{E}_{y_t=x} \left[\int_t^{\tau(x)} \Psi ds + \Phi(y_{\tau(x)}) \right]$ satisfies the same PDE on (a, b) with *boundary conditions*

$$u(a, t) = \Phi(a), \quad u(b, t) = \Phi(b).$$

Both values of Φ at the endpoints must be specified for unique solvability.

1.4 Multi-Dimensional Diffusions

For n assets $y_s^{(i)}$ with SDE

$$dy_s^{(i)} = F_i(y_s, s) ds + \sum_{j=1}^n G_{ij}(y_s, s) dW_s^{(j)},$$

the generator is

$$\mathcal{L} = \sum_{i=1}^n F_i \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i,j,k=1}^n G_{ik} G_{jk} \frac{\partial^2}{\partial x_i \partial x_j},$$

and $u(x, t) = \mathbb{E}_{y_t=x}[\Phi(y_T)]$ solves $\partial_t u + \mathcal{L}u = 0$ with $u(x, T) = \Phi(x)$.

1.5 The Black-Scholes PDE (BSPDE)

For GBM stock $dy_t = \mu y_t dt + \sigma y_t dW_t$ with constant interest rate $b = r$, no-arbitrage pricing gives :

$$\frac{\partial P}{\partial t} + \frac{\sigma^2 x^2}{2} \frac{\partial^2 P}{\partial x^2} + r \left(x \frac{\partial P}{\partial x} - P \right) = 0, \quad P(x, T) = \Phi(x).$$

Derivation (Chapter 1) : via the self-financing portfolio replication argument : matching dW_t -coefficients forces $a_t = \partial_x P$, and matching dt -coefficients gives the above PDE.

2 Linear PDE Theory (Chapter 2)

2.1 First-Order PDE : Transport Equation

Constant-coefficient transport equation :

$$\frac{\partial u}{\partial t} + b \cdot \nabla_x u = 0, \quad u(x, 0) = h(x).$$

Solution :

$$u(x, t) = h(x - tb).$$

Values are transported along straight lines $x + sb, t + s$.

Linear first-order PDE (method of characteristics) :

$$a(x, t) \cdot \nabla_{x,t} u(x, t) + c(x, t) u(x, t) = 0.$$

Find characteristic curve $\gamma(s) = (x(s), t(s))$ satisfying $\dot{\gamma} = a(\gamma)$. Setting $v(s) = u(\gamma(s))$, it follows that

$$\frac{dv}{ds} = c(\gamma(s)) v(s) \quad \implies \quad v(s) = v(0) e^{\int_0^s c(\gamma(r)) dr}.$$

Sheet 2, Q5 example : PDE $\partial_x u + \partial_y u = u$ with $u(x, 0) = f(x)$. Characteristics $\gamma(s) = (s + r, s)$ give $v(s) = e^s f(r)$, so

$$u(x, y) = e^y f(x - y).$$

The boundary condition $u(x, x) = 1$ is prescribed on a characteristic ($r = 0$), forcing $v(s) \equiv 1$, which contradicts $v' = v$: **no solution exists**. For $u(x, x) = 0$, the condition forces $v(s) = 0$ on $r = 0$ but leaves all other lines undetermined : **infinitely many solutions**.

2.2 Laplace and Poisson Equations

For a Brownian motion $dy_s^{(i)} = dW_i(s)$ in a bounded domain D , the expected payout $u(x) = \mathbb{E}_{y_0=x}[\Phi(y_{\tau(x)})]$ satisfies the **Laplace equation** :

$$\Delta u = 0 \text{ on } D, \quad u = \Phi \text{ on } \partial D.$$

Including running payoff f , one obtains the **Poisson equation** :

$$\Delta u + f = 0 \text{ on } D, \quad u = \Phi \text{ on } \partial D.$$

Fundamental solution (Sheet 2, Q2) :

$$F(x) = \begin{cases} -\frac{1}{2\pi} \ln |x| & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}} & n \geq 3, \end{cases}$$

where $\alpha(n)$ is the volume of the unit ball. Verification : compute $\partial^2 F / \partial x_i^2$ and sum to show $\Delta F = 0$.

Green's function method : Solve $\Delta_y \phi_x = 0$ in D with $\phi_x = F(y - x)$ on ∂D . Then $G(x, y) = F(y - x) - \phi_x(y)$, and the Poisson equation is solved by

$$u(x) = \int_{\partial D} \Phi(y) \frac{\partial G}{\partial \nu}(x, y) dS(y) - \int_D f(y) G(x, y) dy.$$

2.3 Time-Independent Operator (Sheet 2, Q3; Sheet 3)

If coefficients f_i, g_{ij} do not depend on t , the expected payout

$$u(x) = \mathbb{E}_{y_0=x} \left[\int_0^{\tau(x)} \Psi(y_s) ds + \Phi(y_{\tau(x)}) \right]$$

satisfies the **elliptic BVP** :

$$\mathcal{L}u + \Psi = 0 \text{ in } D, \quad u = \Phi \text{ on } \partial D,$$

where $\mathcal{L} = \sum_i f_i \partial_{x_i} + \frac{1}{2} \sum_{i,j,k} g_{ik} g_{jk} \partial_{x_i x_j}^2$.

Expected exit time (Sheet 2, Q1) : Setting $\Psi = 1, \Phi = 0$, and taking $t \rightarrow -\infty$ in the time-dependent formulation reduces to the ODE

$$F \frac{\partial u}{\partial x} + \frac{G^2}{2} \frac{\partial^2 u}{\partial x^2} = -1, \quad u(a) = u(b) = 0.$$

For GBM ($F = \mu x, G = \sigma x$) :

$$\mu x \frac{\partial u}{\partial x} + \frac{\sigma^2 x^2}{2} \frac{\partial^2 u}{\partial x^2} = -1, \quad u(a) = u(b) = 0.$$

General solution (for $\mu \neq \frac{1}{2}\sigma^2$) :

$$u(x) = \frac{\ln x}{\sigma^2/2 - \mu} + C_1 + C_2 x^\gamma, \quad \gamma = 1 - \frac{2\mu}{\sigma^2}.$$

Constants C_1, C_2 are determined by $u(a) = u(b) = 0$.

2.4 Scaling Relations (Sheet 2, Q4)

If $u : \mathbb{R}^n \rightarrow \mathbb{R}$ solves $x \cdot \nabla u(x) = \alpha u(x)$, then

$$u(\lambda x) = \lambda^\alpha u(x) \quad \forall \lambda > 0.$$

Technique : Define $v(\lambda) = u(\lambda x)$, derive the ODE $v' = (\alpha/\lambda)v$ via chain rule plus PDE, and solve.

3 Heat Equation (Chapter 2 & Sheets 2-3)

3.1 The Heat Equation and Its Fundamental Solution

$$\frac{\partial u}{\partial t} = \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x).$$

The **fundamental solution** (heat kernel) is

$$\Phi(x, t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{|x|^2}{2\sigma^2 t}\right),$$

and the full-space solution is :

$$u(x, t) = \int_{\mathbb{R}^n} \Phi(x - y, t) u_0(y) dy = \frac{1}{(2\pi\sigma^2 t)^{n/2}} \int_{\mathbb{R}^n} e^{-|x-y|^2/(2\sigma^2 t)} u_0(y) dy.$$

3.2 Discontinuous Initial Data (Sheets 2-3, Q1)

For $u_0(x) = \mathbf{1}_{x>0}$, the solution is

$$u(x, t) = \mathcal{N}\left(\frac{x}{\sqrt{t}}\right), \quad \text{where } \mathcal{N}(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-s^2/2} ds.$$

Derivation : Substitution $z = (x - y)/\sqrt{t}$ in the solution formula.

Key properties of $u(x, t) = \mathcal{N}(x/\sqrt{t})$:

- $\partial u / \partial x = (2\pi t)^{-1/2} e^{-x^2/(2t)}$, maximised at $x = 0$ with value $(2\pi t)^{-1/2} \xrightarrow{t \rightarrow 0} \infty$.
- The region $\partial u / \partial x < \frac{1}{10} \max_x \partial u / \partial x$ corresponds to $|x| \geq \sqrt{2t \ln 10} \approx 4.6\sqrt{t}$.

Integrated solution. $v(x, t) = \int_{-\infty}^x u(z, t) dz$ also solves the heat equation with initial condition $v(x, 0) = \max\{0, x\}$. Moreover, $v(0, t) = \sqrt{2t/\pi}$.

3.3 Deriving the Fundamental Solution (Sheet Extra, Q2)

Scaling argument : If u solves the heat equation, so does $u(cx, c^2t)$. Seek self-similar solution $u(x, t) = U(|x|^2/t)$; this satisfies

∂u/∂t = Δu ⇐⇒ 4zU''(z) + (2n + z)U'(z) = 0, z = |x|^2/t.

The general solution is $U(z) = a \int_0^z e^{-s/4} s^{-n/2} ds + b$. In $n = 1$, differentiating $U(x^2/t)$ with respect to x and choosing $a = 1/(4\sqrt{\pi})$ recovers $\Phi(x, t) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$.

3.4 From Black-Scholes to Heat Equation

Change of variables. Put $y = \ln x, \tau = T - t$, and define $u(y, \tau) = P(e^y, T - \tau) e^{-\alpha y - \beta \tau}$,

with

α = 1/2 - b/σ², β = (1/2σ² - 1) b - 1/4.

Then u satisfies

∂u/∂τ - σ²/2 ∂²u/∂y² = 0, u(y, 0) = e^{-αy} Φ(e^y).

Inversion : $P(x, t) = e^{\alpha y + \beta \tau} u(y, \tau)$ with $y = \ln x, \tau = T - t$.

3.5 Heat Equation on a Half-Space : Reflection Trick

To solve $\partial_t u = \frac{\sigma^2}{2} \partial_{xx} u$ for $x > 0, t > 0$ with $u(x, 0) = g(x), u(0, t) = 0$:

Reflection trick : extend the initial condition as $\tilde{g}(x) = g(x) \mathbf{1}_{x>0} - g(-x) \mathbf{1}_{x<0}$. Solve the whole-space problem with $\tilde{g}$:

v(x, t) = ∫_{-∞}^∞ Φ(x - y, t) g̃(y) dy,

then $v(x, t) = \int_0^\infty G(x, y, t) g(y) dy$ with $G(x, y, t) = \Phi(x - y, t) - \Phi(x + y, t)$.

3.6 Barrier Option Pricing (Sheets 2-3, Q2-Q3)

Setup. Knockout call with strike K , knock-out level $\bar{K}$: $\Phi(x) = (x - K)^+$, $P(\bar{K}, t) = 0$.

Step 1 : Transform. After $y = \ln x, \tau = T - t$, the heat equation initial condition becomes $u_0(y) = e^{-\alpha y} (e^y - K)^+$.

Step 2 : Reflect. The zero boundary condition at $y = \ln \bar{K}$ is enforced by reflecting the initial condition :

f0(y) = u0(y) - u0(2 ln K̄ - y).

Reflection formula : $y' = 2 \ln \bar{K} - y$ (same distance from $\ln \bar{K}$).

Step 3 : Invert. Let $u_1(y, \tau), u_2(y, \tau)$ solve the full-space heat equation with $u_0(y)$ and $u_0(2 \ln \bar{K} - y)$ respectively. Then

P(x, t) = C(x, t) - (x/K̄)²α C(K̄²/x, t),

where $C(x, t)$ is the standard vanilla European call price.

3.7 Half-Space Problems and Interpretation (Sheets 2-3, Q3)

w(0, t)	w(x, 0)	Financial interpretation
0	1	Binary : pays 1 if asset stays positive
0	(x - K)⁺, K > 0	Knockout call, barrier at 0
0	(x - K)⁺, K < 0	Same ; steep behaviour near (0, 0)
1	0	Digital : pays 1 if asset crosses zero

Solution technique for all cases : reflection $w(x, t) = u(x, t) - u(-x, t)$ or $w(x, t) = u(-x, t) - u(x, t) + 1$ (case (d)).

3.8 Fourier Series on an Interval (Sheet Extra, Q1)

For $\partial_t u = \partial_{xx} u$ on $[0, 1]$ with $u(0, t) = u(1, t) = 0, u(x, 0) = g(x)$:

u(x, t) = ∑_{k=1}^∞ g_k e^{-½k²π²t} sin(kπx), g_k = 2 ∫_0^1 g(y) sin(kπy) dy.

Method : substitute $u = \sum a_k(t) \sin(k\pi x)$; orthogonality forces $a'_k = -\frac{1}{2} k^2 \pi^2 a_k$, giving $a_k(t) = a_k(0) e^{-k^2 \pi^2 t/2}$.

For $g \equiv 1$, the Fourier coefficients are $g_k = 0$ (even k) and $g_k = 4/(k\pi)$ (odd k), giving

u(x, t) = ∑_{k=1}^∞ 4/(2k-1)π e^{-(2k-1)²π²t} sin((2k-1)πx).

Truncation : at $t = 10^{-2}$, only 3 terms are needed for accuracy 10^{-2} .

3.9 Energy Methods and Uniqueness (Sheet Extra, Q3)

Define the L^2 -energy $E(t) = \int_D u(x, t)^2 dx$.

Key identity (integration by parts) :

dE/dt = 2 ∫_D u Δu dx = -2 ∫_D |∇u|² dx + 2 ∫_{∂D} u ∇u · n dS.

If $u = 0$ on ∂D (Dirichlet), then $dE/dt \leq 0$: energy is non-increasing.

Uniqueness proof : Let $w = u - v$ be the difference of two solutions with equal data. Then E_w is non-increasing and $E_w(0) = 0$, so $E_w(t) = 0$ and $w \equiv 0$.

Growth estimate. For $\partial_t u = \Delta u + \lambda u$ ($\lambda > 0$), $u = 0$ on ∂D :

dE/dt ≤ 2λE(t) ⇒ E(t) ≤ E(0) e^{2λt} (Gronwall's inequality).

Gronwall's Lemma : If $f'(t) \leq af(t)$, then $f(t) \leq f(0)e^{at}$.

3.10 Maximum Principle

Let $D \subset \mathbb{R}^n$ be bounded open. If $\partial_t f \leq \Delta f$ on $D \times (0, T)$, then the maximum of f on $\bar{D} \times [0, T]$ is attained either on $\partial D \times [0, T]$ or on $D \times \{0\}$.

Consequence : Uniqueness for the heat equation on bounded domains.

4 Optimal Control (Chapter 3 & Sheets 4-5)

4.1 Dynamic Programming Principle

State equation :

dy/ds(s) = F(y(s), α(s)), y(t) = x.

Optimal value function :

u(x, t) = max_{α ∈ A} { ∫_t^T h(y(s), α(s)) ds + g(y(T)) }.

Dynamic programming principle :

u(x, t) = max_{α ∈ A} { ∫_t^{t1} h(y(s), α(s)) ds + u(y(t1), t1) }, t1 > t.

4.2 Hamilton-Jacobi-Bellman (HJB) Equation (Sheet 4, Q2)

∂u/∂t(x, t) + max_{α ∈ A} [∇_x u(x, t) · F(x, α) + h(x, α)] = 0, u(x, T) = g(x).

The Hamiltonian is

H(x, p) = max_{α ∈ A} [p · F(x, α) + h(x, α)].

4.3 HJB with Discounting (Sheet 4, Q1&2)

For $v(x, t) = \max_\alpha \left\{ \int_t^T e^{-\rho(s-t)} h(y(s), \alpha(s)) ds + e^{-\rho(T-t)} g(y(T)) \right\}$:

∂v/∂t - ρv + H(x, ∇v) = 0, v(x, T) = g(x).

Technique : write $v = e^{\rho t} u$; the $e^{-\rho t}$ factor produces the $-\rho v$ term.

Infinite horizon (Sheet 4, Q2b). $\tilde{v}(x) = \max_\alpha \int_t^\infty e^{-\rho(s-t)} h(y(s), \alpha(s)) ds$ is independent of t and satisfies the stationary HJB equation :

-ρṽ + H(x, ∇ṽ) = 0.

4.4 Optimal Consumption Problem (Sheet 4, Q1)

Setup : $dy/ds = ry(s) - \alpha(s)$, utility $u(x, t) = \max_\alpha \int_t^T e^{-\rho s} \alpha(s)^\gamma ds, 0 < \gamma < 1$.

Scaling property (Step (a)) :

u(λx, t) = λ^γ u(x, t).

Proof : for $y(t) = \lambda x$, use control $\lambda \alpha_x(s)$; the constraint $y \geq 0$ is preserved and the integral scales by λ^γ .

HJB equation (Step (b)) :

∂u/∂t + max_{α ≥ 0} [(rx - α) ∂u/∂x + e^{-ρt} α^γ] = 0, u(x, T) = 0.

$u(x, t) = g(t)x^\gamma$ (forced by scaling). Differentiating the HJB max w.r.t. α :

α_max = (g(t)e^{ρt})^{1/(γ-1)} x.

Substituting back yields the ODE for g :

dg/dt + rγg + (1 - γ)(e^{ρt}g)^{1/(γ-1)}g = 0, g(T) = 0.

Set $G = e^{\rho t} g$, then $H = G^{1/(1-\gamma)}$ satisfies the linear ODE :

dH/dt + μH + 1 = 0, H(T) = 0, μ = (rγ - ρ)/(1 - γ).

Solution (Step (d)) :

H(t) = 1/μ (e^{μ(T-t)} - 1), g(t) = e^{-ρt} ((rγ - ρ)/(1 - γ))^{γ-1} (exp((rγ - ρ)/(1 - γ) (T - t)) - 1)^{1-γ},

and the optimal control is

α*(x, t) = (rγ - ρ)/(1 - γ) (exp((rγ - ρ)/(1 - γ) (T - t)) - 1)^{-1} x.

Approximation for small r, ρ : $\alpha^*(x, t) \approx x/(T - t)$ (constant consumption rate).

4.5 Stochastic Optimal Control (Lecture 3 & Sheet 5)

State SDE :

dy_s = f(y_s, α_s) ds + σ(y_s, α_s) dW_s, y_t = x.

Stochastic HJB equation :

∂u/∂t + max_{α ∈ A} { ∇u · f(x, α) + h(x, α) + σ²(x, α)/2 Δu } = 0, u(x, T) = g(x).

4.6 Merton's Portfolio Problem (Lecture 3 & Sheet 5)

Setup : Bond $dq_s = r q_s ds$ and stock $dp_s = \mu p_s ds + \sigma p_s dW_s$. Let $\alpha(s)$ = fraction of wealth in stock. Portfolio wealth :

dy_s = α(s)y_s(μ ds + σ dW_s) + (1 - α(s))y_s r ds = y_s [r + α(s)(μ - r)] ds + α(s)σ y_s dW_s.

Objective : $u(x, t) = \max_\alpha \mathbb{E}_{y_t=x} [h(y_T)]$ with $h(y) = y^\gamma, 0 < \gamma < 1$.

HJB equation :

∂u/∂t + max_α { ∂u/∂x (rx + α(μ - r)x) + σ²α²x²/2 ∂²u/∂x² } = 0.

Optimal allocation :

α* = - (μ - r) ∂_x u / (σ² x ∂²_x u).

Ansatz $u(x, t) = g(t)x^\gamma, g(T) = 1$. The HJB becomes :

g'(t) + (rγ + λ²γ/(2(1 - γ))) g(t) = 0, λ = (μ - r)/σ.

Solution :

$$g(t) = \exp\left(-\left(r\gamma + \frac{\lambda^2\gamma}{2(1-\gamma)}\right)(t-T)\right), \quad \alpha^* = \frac{\mu-r}{(1-\gamma)\sigma^2} \quad (\text{constant}).$$

The optimal stock fraction is independent of wealth and time. It is valid when $0 \leq \mu - r \leq \sigma^2(1 - \gamma)$.

5 Numerical Methods (Chapter 4 & Sheet 5)

5.1 Forward (Explicit) Euler for ODE

For $y'(t) = f(y(t), t)$, $y(t_0) = y_0$:

$$z_{j+1} = z_j + h\,f(z_j, t_j).$$

Truncation error : $\varepsilon_j^h = O(h)$ (first-order consistent).

5.2 Second-Order Taylor Scheme (Sheet 5, Q2a)

Expand $y(t_j + h)$ to second order and use $y' = f$, $y'' = (\partial_y f)f + \partial_t f$:

$$\begin{aligned} z_{j+1} = & z_j + h\,f(z_j, t_j) \\ & + \frac{h^2}{2}\left(\frac{\partial f}{\partial y}(z_j, t_j)f(z_j, t_j) + \frac{\partial f}{\partial t}(z_j, t_j)\right) + O(h^3). \end{aligned}$$

5.3 Derivative-Free Second-Order Scheme (Sheet 5, Q2b)

Define $\varphi(h) = f(z_j + hf(z_j, t_j), t_j + h)$. First-order Taylor :

$$\varphi(h) = f(z_j, t_j) + h(\partial_y f \cdot f + \partial_t f) + O(h^2).$$

Substituting into the scheme of Q2a, the derivative terms combine :

$$z_{j+1} = z_j + \frac{h}{2}(\xi_{j-1} + f(z_j + h\xi_{j-1}, t_j)), \quad \xi_{j-1} = f(z_{j-1}, t_{j-1}).$$

This is a **Runge-Kutta type** (Heun’s method) second-order scheme requiring only evaluations of f .

5.4 Convergence, Consistency, Stability

Truncation error :

$$\varpi_j^h = y(t_{j+1}) - y(t_j) - h\,\phi^h(y(t_j), t_j).$$

A scheme is *consistent of order p* if $\max_j |\varpi_j^h| = O(h^p)$.

Convergence from Lipschitz continuity : If ϕ^h is Lipschitz in y (constant L) and the truncation error is $O(h^p)$, then

$$\max_j |z_j - y(t_j)| \leq \frac{M}{L}\big(e^{(T-t_0)L} - 1\big)h^p,$$

i.e. the scheme is convergent of order p .

5.5 Discrete Gronwall Lemma (Sheet Extra, Q5)

If $e_j \leq Ce_{j-1} + D$ for $C > 1$, $D > 0$, $j \geq 1$, then

$$e_j \leq D\,\frac{C^j - 1}{C - 1} + e_0\,C^j.$$

Proof : induction on j . When $C = 1$: $e_j \leq jD + e_0$.

5.6 Numerical PDE : Forward Scheme for Heat Equation

Space-time grid : $x_i = a + h_x i$, $t_j = t_0 + h_t j$. Discretise :

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_j) \approx \frac{u(x_{i+1}, t_j) + u(x_{i-1}, t_j) - 2u(x_i, t_j)}{h_x^2}.$$

Explicit (forward) scheme :

$$v_i^{j+1} = v_i^j + \mu\big(v_{i+1}^j + v_{i-1}^j - 2v_i^j\big), \quad \mu = \frac{\sigma^2 h_t}{2h_x^2}.$$

Matrix form : $\mathbf{v}^{j+1} = (I + \mu A)\mathbf{v}^j$, where A is the **discrete Laplacian**.

Eigenvalues of A : $(w_k)_i = \sin(k\pi i/M)$, with $\rho_k = -2(1 - \cos(k\pi/M))$.

Stability condition : the scheme is stable if and only if

$$|1 + \mu\rho_k| < 1 \text{ for all } k \quad \Longleftrightarrow \quad \mu \leq \tfrac{1}{2}.$$

If $\mu > 1/2$, the scheme diverges (oscillations grow).

Implicit (backward) scheme :

$$v_i^j = v_i^{j-1} - \mu\big(v_{i+1}^j + v_{i-1}^j - 2v_i^j\big), \quad \text{i.e.} \quad \mathbf{v}^j = (I - \mu A)^{-1}\mathbf{v}^{j-1}.$$

Eigenvalues of $(I - \mu A)^{-1}$ are $(1 + 2\mu(1 - \cos(k\pi/M)))^{-1} < 1$ for all $\mu > 0$: **unconditionally stable**.

5.7 The Lax Equivalence Theorem

For a linear, well-posed PDE, a consistent finite-difference scheme is *convergent* if and only if it is *stable*.

- Forward Euler : consistent, but stable only for $\mu \leq \frac{1}{2}$ (conditionally stable).
- Backward Euler : consistent and unconditionally stable $\Rightarrow$ always convergent.

6 Interconnections and Summary

Concept	Key formula / result	Problem Sheet
Itô’s formula	$du = (\partial_t u + F\partial_x u + \frac{\sigma^2}{2}\partial_{xx}u)\,dt + G\partial_x u\,dW$	All PDE derivations (Ch. 1)
Kolmogorov backward eq.	$\partial_t u + \mathcal{L}u = 0$	Option pricing, knockout options (Ch. 1, Sh. 1)
Black-Scholes PDE	$\partial_t P + \frac{\sigma^2 x^2}{2}\partial_{xx}P + r(x\partial_x P - P) = 0$	Barrier options (Ch. 2, Sh. 2-3)
BS $\rightarrow$ heat transform	$y = \ln x, \tau = T - t, u = Pe^{-\alpha y - \beta \tau}$	Barrier option pricing (Ch. 2, Sh. 2-3)
Reflection trick	$\tilde{g}(x) = g(x) - g(-x)$	Half-space heat eq., barrier pricing (Ch. 2)
Fundamental solution (heat)	$(4\pi t)^{-1/2}e^{-x^2/(4t)}$	Heat equation, option formulas (Ch. 2, Sh. Extra)
Fourier series	$u = \sum a_k(t)\sin(k\pi x), \quad a'_k = -\frac{1}{2}k^2\pi^2 a_k$	Double barrier options (Ch. 2, Sh. Extra)
Max principle / uniqueness	$\max f$ on boundary ; energy $\rightarrow 0$	Well-posedness (Ch. 2, Sh. Extra)
HJB equation (det.)	$\partial_t u + \max_\alpha [\nabla u \cdot F + h] = 0$	Optimal consumption (Ch. 3, Sh. 4)
HJB with discounting	$\partial_t v - \rho v + H(x, \nabla v) = 0$	Discounted control (Ch. 3, Sh. 4)
Stochastic HJB	$\partial_t u + \max_\alpha [\nabla u \cdot f + h + \frac{\sigma^2}{2}\Delta u] = 0$	Merton problem (Ch. 3, Sh. 5)
Merton optimal allocation	$\alpha^* = (\mu - r)/((1 - \gamma)\sigma^2)$: constant	Portfolio selection (Ch. 3, Sh. 5)
Forward Euler (ODE/PDE)	$z_{j+1} = z_j + hf(z_j, t_j)$; stable for $\mu \leq \frac{1}{2}$	Numerical PDE (Ch. 4, Sh. 5)
Backward Euler (PDE)	$(I - \mu A)\mathbf{v}^j = \mathbf{v}^{j-1}$; unconditionally stable	Numerical PDE (Ch. 4)
Lax theorem	Consistent + stable $\Leftrightarrow$ convergent (linear PDE)	Numerical analysis (Ch. 4)
Discrete Gronwall	$e_j \leq D\frac{C^j-1}{C-1} + e_0C^j$	Error estimates (Ch. 4, Sh. Extra)